

OM1: Practice Problems on CPM/Project Crashing and Inventory Build-Up Diagram

1. **WIMWI Studios** is run by SoChe Kumar, a WIMWI alumnus, who hated the Operations Management courses during his WIMWI days because of the deadly quizzes. Now, however, he applies most of the stuff he picked in those courses. At the moment, WIMWI studios is about to begin the production of its most important (and most expensive) movie of the year. SoChe has decided to use the project management concepts to help plan and control this key project. He has identified the eight major activities and their precedence relationship as given in Table 1. SoChe, however, has gathered that another studio, WIMSI, also will be coming out with a blockbuster movie during the middle of the upcoming summer, just when his movie is to be released. Therefore, he and his top management team have concluded that they must accelerate production of the movie and bring it out at the beginning of the summer (15 weeks from now) to establish it as *the movie* of the year. Although this will require substantially increasing an already huge budget, management feels that this will pay off in much larger box office earnings, both nationally and internationally. SoChe has also gathered additional data as given in Table 2. Assume the cost of crashing any activity per week is constant.

Determine the least costly way of meeting the new deadline of 15 weeks and the corresponding additional cost. **Ans: 217 Crores**

Activity	Immediate Predecessor	Normal Duration (Weeks)
A	-	5
B	-	3
C	A	4
D	A	6
E	B	5
F	B	7
G	C,E	9
H	D,F	8

Table 1: Activity precedence and normal time

Activity	Crash Time (Weeks)	Normal Cost (Crores)	Crash Cost (Crores)
A	3	20	30
B	2	10	20
C	2	16	24
D	3	25	43
E	4	22	30
F	4	30	48
G	5	25	45
H	6	30	44

Table 2: Crash times and costs

2. The number of devotees visiting and offering prayers at Shri Mata Vaishno Devi Holy Cave, a popular pilgrimage centre in North India, has been steadily increasing over time, with approximately 9 million visiting in 2014 (the number is likely to be more than 10 million in 2015). The immense popularity of the Shrine and its topography poses significant challenges to the management of the waiting lines and waiting times for the devotees. In order to implement effective crowd management system so as to take care of ever growing devotees and their waiting experience, IIM Ahmedabad students are invited (through this question!) to critically review the existing system. Shri Mata Vaishno Devi Shrine Board, which manages the shrine, has provided the following data.
- The Holy Cave is open to Devotees for *darshan* (to offer prayers) around the clock, except during 06:00-08:00 and 18:00-20:00 hours when *aarti* (religious ritual of worship) is performed by the priests. About 2,400 devotees can have *darshan* every hour at the Holy Cave. On a typical peak day, about 44,000 devotees visit the cave. However, it has been observed that number of devotees arriving to visit the cave varies during the day. Table 1 shows the devotee arrival pattern. You can ignore the intra-hour variations.

Table 1: Devotee Arrival Pattern

Hours	Arrivals/ hour
0600-1000	2000
1000-1600	1500
1600-2000	2500
2000-0200	2000
0200-0600	1250

- What is the maximum queue length that one will observe? **Ans: 5,200 (by the end of 20:00 hrs)**
- What is the average waiting time for the devotees before *darshan*? **Ans: 68.38 min**
- Shri Mata Vaishno Devi Shrine Board feels that very long queues put too much stress on employees, security agencies, administrators and volunteers etc. They have decided to influence the arrival pattern by simply putting a statement on their website about possibilities of a quick *darshan* during afternoon hours. Experts believe that the arrival pattern would change (although there may not be any change in total devotees visiting the shrine). The devotee arrival pattern is likely to be as shown in table 2.

Table 2: Influenced Devotee Arrival Pattern

Hours	Arrivals/ hour
0600-1000	1700
1000-1600	2000
1600-2000	2050
2000-0200	2000
0200-0600	1250

- What is the maximum queue length that one will observe now? **Ans: 4,100 (by the end of 20:00 hrs)**
- Now, what will be the average waiting time for the devotees before *darshan*? **Ans: 50.045 min**